Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms. The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U, f)], [I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 ([8, Corollary 10.3, Proposition 10.10(1)])**.** *Ψ and $\mu\Psi$ are both isomorphisms of graded semirings.*

Formalization review T001: the measured kernel. There is a measured counterpart to the preceding kernel calculation. It is nontrivial over the same original ground, but the translated Gamma volume form must be corrected. Put

$$T_\mu = \mathbf{K}\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}\,\mu\Gamma[*].$$

This is a mathematical calculation; the complete geometric kernel proof has not yet been exported in Lean.

Let $\boldsymbol{X} = (X, f, \omega) \in \mu\mathrm{RES}[r]$, let $\delta \in \Gamma$ be definable, and suppose there is a definable function $c : X \longrightarrow \delta^\sharp$, all over the original ground $\mathbb{S}$. Take $(U, \sigma) \in \mu\Gamma[m]$, $m \geq 1$, and a coordinate vector e_i. Set

$$U_\delta = U + \delta e_i, \qquad \sigma_\delta(\eta) = \sigma(\eta - \delta e_i) - \delta.$$

Then the homogeneous generating presentation is

$$\boxed{\ker(\mu\Psi) = \Big([\boldsymbol{X}]_r \otimes \big([(U,\sigma)]_m - [(U_\delta, \sigma_\delta)]_m\big) : \boldsymbol{X}, c, U, \sigma, i \text{ as above}\Big).}$$

In the positive semiring this means the congruence generated by the corresponding pairs. No minimal generating set is claimed.

The sign and the generating bijection. The map multiplying the ith free RV coordinate by $c(x)$ has Γ-Jacobian δ. The target form is smaller by δ, so the measured morphism condition is satisfied. Equivalently, put $\lambda_U(\gamma) = \sigma(\gamma) + \Sigma\gamma$. Then

$$\sigma_\delta(\gamma + \delta e_i) + \Sigma(\gamma + \delta e_i) = \sigma(\gamma) + \Sigma\gamma.$$

Thus the *combined weight* is simply transported by the translation. The form on $\boldsymbol{X}$ is unchanged; no RV point of $\delta^\sharp$ has been chosen over $\mathbb{S}$ alone.

Why the generators are complete. Use the original-ground local form $F(x,u) = (h(x), c(x)u^M)$, with $M \in \mathrm{GL}_m(\mathbb{Z})$, $\mathrm{vrv}(c(x)) = \boldsymbol{\delta}$ constant, and $k(\gamma) = M\gamma + \boldsymbol{\delta}$. If A, B are the combined RES weights and λ, ν the combined Gamma weights, the measured condition is

$$A(x) + \lambda(\gamma) = B(h(x)) + \nu(k(\gamma)).$$

On a nonempty product rectangle both sides separate, hence

$$A(x) - B(h(x)) = \nu(k(\gamma)) - \lambda(\gamma) = a$$

for one original-ground definable constant a. The degree-zero finite measured Gamma coefficient $p_a = [(\{*\}, a)]_0$ shifts a volume form by a; it is invertible, with inverse p_{-a}. Shift the target RES form by a, making h measured, and balance p_a to the Gamma factor. The unimodular part then transports its combined weight by an existing Gamma morphism; the displayed generators handle the translations one coordinate at a time. Compactness gives finitely many disjoint matched rectangles. Fixed Gamma coordinates, with compensating forms, are absorbed into RES by finite-Gamma balancing. Induction on the remaining Gamma rank and positive scissors complete the proof. This uses no cancellation and retains the RES-dependent multipliers.

An explicit nonzero element. Over $\mathbb{S} = \mathbb{C}((t))$, $\mathrm{val}(t) = 1$, put $R = \{r : r^5 = \mathrm{rv}(t)\}$, $\delta = 1/5$, $I = (0, \infty)$ and $J = (\delta, \infty)$. Then

$$\Delta_\mu = [(R,0)]_0 \otimes \big([(I,0)]_1 - [(J,-\delta)]_1\big) \neq 0, \qquad \mu\Psi(\Delta_\mu) = 0.$$

The map $(r,u) \mapsto (r, ru)$ is measured with these forms: $0 = (-\delta) + \delta$. Forgetting volume forms sends Δ_μ to the previously detected nonzero unmeasured element, proving its nonvanishing even after groupification.

More generally, giving any unmeasured object (X, f) the form $-\Sigma(\mathrm{vrv} \circ f)$ makes its combined weight zero. This construction respects products, scissors and all four tensor categories, and is a section of forgetting the forms. It embeds the unmeasured kernel into the measured kernel. In the example this gives the alternative generator $[(R,0)]_0 \otimes \big([(I,-\mathrm{id})]_1 - [(J,-\mathrm{id})]_1\big)$.

The same degree-one annihilation. With

$$\tau_\mu = 1 \otimes [(\{0\}, 0)]_1 = [(\mathbb{G}_m, \mathrm{id}, 0)]_1 \otimes 1, \qquad \tau_\mu \ker(\mu\Psi) = 0.$$

Use the measured RES bijection $\boldsymbol{X} \times (0^\sharp, \mathrm{id}, 0) \simeq \boldsymbol{X} \times (\delta^\sharp, \mathrm{id}, -\delta)$ given by $c(x)$. Balance the latter finite measured coefficient to the Gamma side and apply the integral shear $(\alpha, \gamma) \mapsto (\alpha, \gamma + \alpha e_i)$ on $\alpha = \delta$, with the shifted form above. Balance back to obtain annihilation of each generator. Consequently $\mu\Psi[\tau_\mu^{-1}]$ is an isomorphism. The element τ_μ has degree one and is not cancelled in the original graded ring. No equality with its entire annihilator is asserted.

The complete mathematical argument is recorded in the section's measured-kernel verification note. The uniform compactness/scissor argument and complete concrete counterexample remain to be finished in Lean. No HK theorem has been used as a premise, and no change has been incorporated into the working manuscript.

Note that, however, [8, Proposition 10.10(2)] does not hold. One of the main motivations of this paper is to construct an alternative to be used in applications.

Notation 3.19. For simplicity, we often drop the constant volume form 0 from the notation. For instance, if $\boldsymbol{U}$ is an object of $\mathrm{RV}[*]$ then it may also denote the object $(\boldsymbol{U}, 0)$ of $\mu\mathrm{RV}[*]$ with the constant volume form 0.

Notation 3.20. Let $[1] \in \mathbf{K}_+\,\mathrm{RES}[1]$ be the class of the singleton $\{1\}$. The class of the singleton $\{1\}$ in $\mathbf{K}_+\,\mathrm{RES}[0]$ is the multiplicative identity of $\mathbf{K}_+\,\mathrm{RES}[*]$ and hence is simply denoted by 1. We have $[\mathrm{RV}_\infty^{\circ\circ}] = [\mathrm{RV}^{\circ\circ}] + 1$ in $\mathbf{K}_+\,\mathrm{RV}[\leq 1]$.

Let I_{sp} be the (nonhomogeneous) semiring congruence relation on $\mathbf{K}_+\,\mathrm{RV}[*]$ generated by the pair $([1], [\mathrm{RV}_\infty^{\circ\circ}])$. Let

$$\boldsymbol{P} = [1] - [\mathrm{RV}^{\circ\circ}] \in \mathbf{K}\,\mathrm{RV}[1].$$

The corresponding principal ideal of $\mathbf{K}\,\mathrm{RV}[*]$ is thus generated by the element $\boldsymbol{P} - 1$.

Similarly, let $\mu\mathrm{I}_{\mathrm{sp}}$ be the semiring congruence relation on $\mathbf{K}_+\,\mu\mathrm{RV}[*]$ generated by the pair $([1], [\mathrm{RV}^{\circ\circ}])$, which is homogeneous, and the corresponding principal ideal of $\mathbf{K}\,\mu\mathrm{RV}[*]$ is generated by the element $\boldsymbol{P}$.

Resolved referee R051 [Typo] – Spelling of homogeneous.
Original source line 1358. In this notation, replace "nonhomogenous" and "homogenous" by "nonhomogeneous" and "homogeneous," respectively.

Author resolution. The author approves correcting the spellings to nonhomogeneous and homogeneous. The congruences and their generators are unchanged.

Notation 3.21. For each $\boldsymbol{U} = (U, f) \in \mathrm{RV}[k]$, let U_f be the set $\bigcup\{f(u)^\sharp \times u \mid u \in U\}$. Let $\mathbb{L}_{\leq k} : \mathrm{RV}[\leq k] \longrightarrow \mathrm{VF}[k]$ be the map given by $\boldsymbol{U} \longmapsto U_f$. Set $\mathbb{L} = \bigcup_k \mathbb{L}_{\leq k}$.

Let $\mu\mathbb{L}_k : \mu\mathrm{RV}[k] \longrightarrow \mu\mathrm{VF}[k]$ be the map given by $(\boldsymbol{U}, \omega) \longmapsto (\mathbb{L}\boldsymbol{U}, \mathbb{L}\omega)$, where $\mathbb{L}\omega$ is the obvious function on $\mathbb{L}\boldsymbol{U}$ induced by ω. Set $\mu\mathbb{L} = \bigoplus_k \mu\mathbb{L}_k$.

Suppose that $\mathbb{S}$ is (VF, Γ)-generated. The map $\mathbb{L}$ induces a surjective semiring homomorphism $\mathbf{K}_+\,\mathrm{RV}[*] \longrightarrow \mathbf{K}_+\,\mathrm{VF}_*$ and the map $\mu\mathbb{L}$ induces a surjective graded semiring homomorphism $\mathbf{K}_+\,\mu\mathrm{RV}[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{VF}[*]$, see [8, § 4, § 6] or [17, Corollaries 7.7, 10.6]; we use the same notation for these homomorphisms as well as their groupifications. We have

Resolved referee R053 [Error; revised] – Scope of the parameter hypothesis.
Original source line 1366. State here the (VF, Γ)-generation hypothesis used in Theorem 3.22, before asserting that lifting induces these homomorphisms. Without a parameter restriction, even well-definedness can fail. For example, let the parameters be generated by a trivially valued algebraically closed field C and a named $\tau \in \mathrm{RV}$ of positive value. The singleton objects $\{\tau\}$ and $\{1\}$ are isomorphic in $\mathrm{RV}[1]$. But $\tau^\sharp$ has no definable VF point: Lemma 2.13 places every such point in $\mathrm{acl}(C) \cap \mathrm{VF} = C$. In contrast, $1 + \mathcal{M}$ contains the definable point 1, so their lifts cannot be definably bijective. Moving the stated hypothesis forward removes this obstruction.